# Database Management Systems

# CPS 3740

**Lecture 1**

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# About the Instructor

# Why do you choose this course?

- **CPS 3740 is NOT required for CS**

  Why Learning Database?
- DB is one of the most important subject in CS.
- The course is designed to help you get

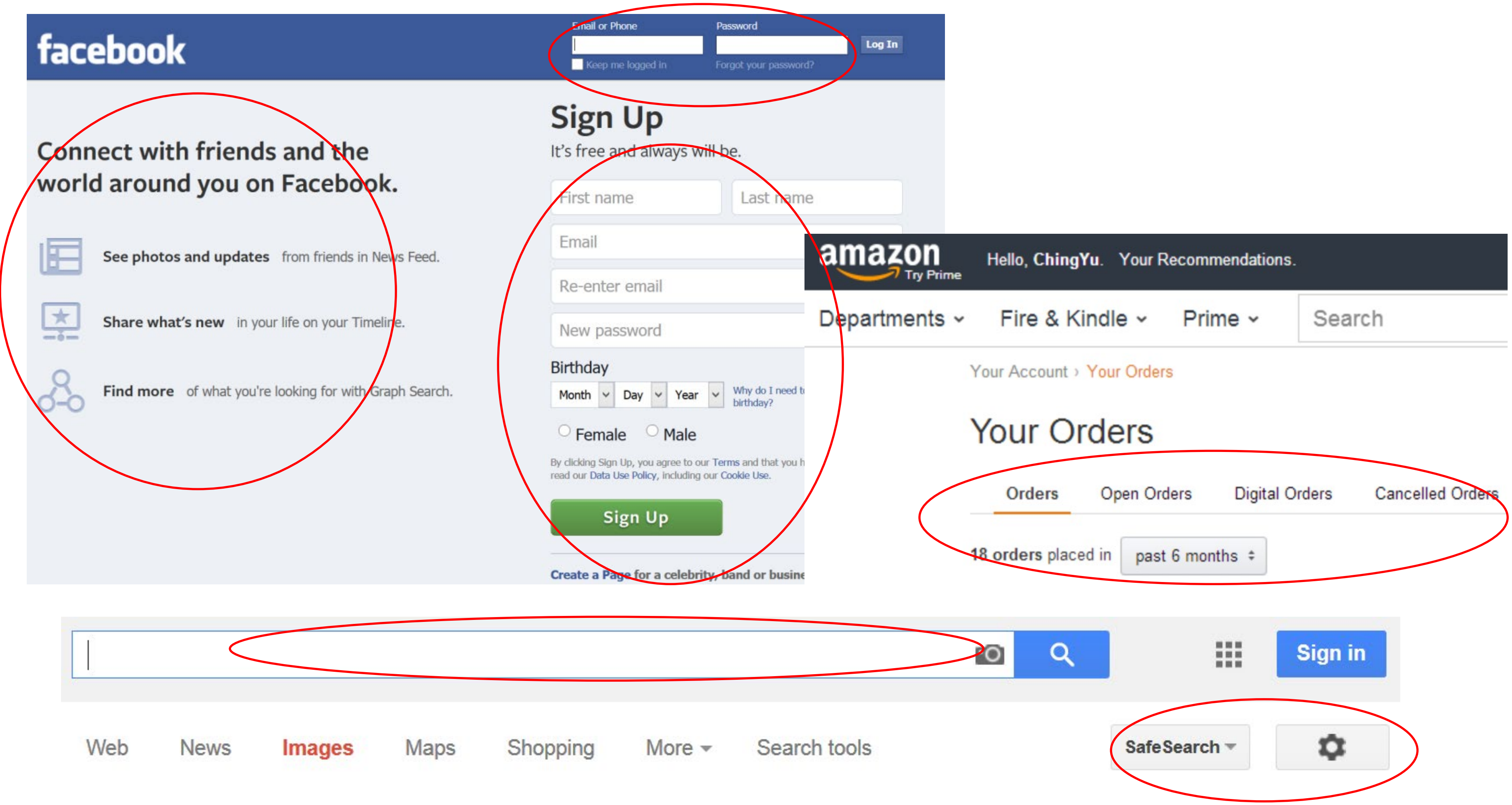
  internship and job.
- Do the assignments by yourself.
- You should list the project on your resume.

# What is "Database"

- A database is an organized collection of data.
  - Data storage & access
  - Data relationship
  - Data and user permission management
  - Transaction management
  - Backup and restore data
- We focus on
  - Only basic database knowledge
  - Web database application design
- Teaching and learning
  - Class discussions
  - A lot of self-study and exercises.

# Applications vs. Information

- What do you see from these pictures?



# Online Store – Amazon.com

- Needs large commercial databases
- Have enormous records
- Store many information (add, change, update)
  - Products (text, images, audio, video)
  - Seller, buyer (name, contact, items)
  - Transaction (order, payment) -> graduate
  - Relationship between information
- Security & protection -> IT, CS, graduate
- Management, backup & maintenance -> IT
- Data mining, prediction, personalization -> CS

- Without keeping the important information, many applications are useless.
- Where are the information stored?

## Learning Database is NOT only

- How to create tables

## We will learn in this class :

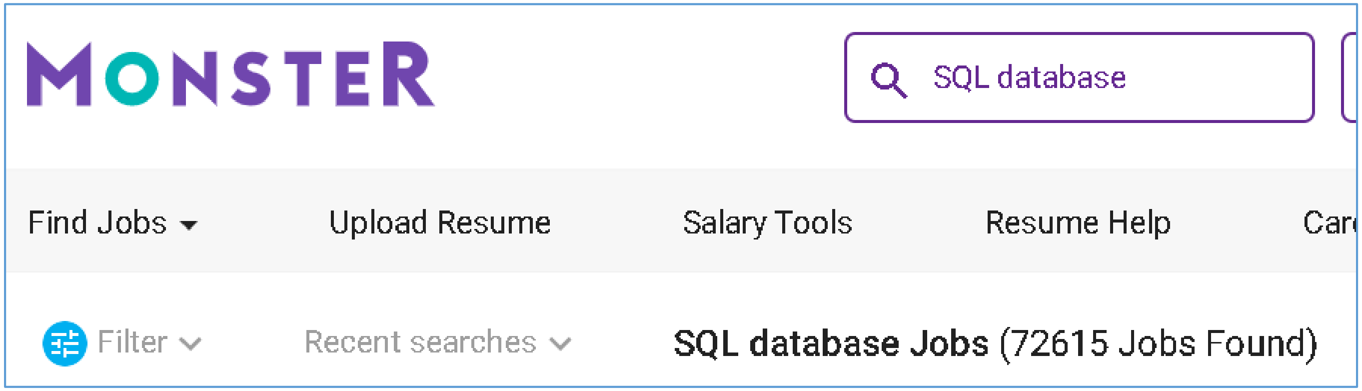
- How to create better tables
- How to store and retrieve the data
- Web database application development

# Career Related to Database

- Software application developer (Apps, Web, Standalone)
  - Programming languages, SQL , business knowledge
- Database administrator (DBA)
  - SQL programming, DBMS, OS
- Business database (data) analyst
  - Business knowledge, SQL
- Search “SQL”, “PL/SQL”, “database developer”, “DBA”
  - www.monster.com
- Database developer salary
  - https://www.indeed.com/career/database-developer/salaries

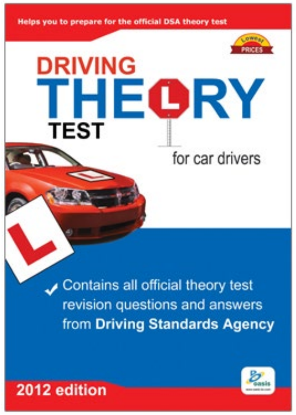
# Plan ahead

- You should be looking for job/internship NOW!
    - You should have sent (or sending) out many resumes.
- You might miss class for job/internship search
    - Phone interview, on-site interview
    - Try to arrange online test at home
- Put the project on your resume, e-portfolio
    - You might use the class for job search/interviews

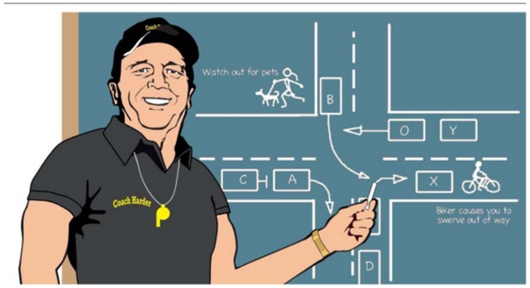


# Knowledge != Skills

- Corporates are looking for skills, NOT just knowledge.
- In computer science, only “hands-on” can transfer knowledge to skills -> Experiences.
- This is a “Hands-on” class:
  - MUST DO exercises and assignments
  - PHP & SQL: a lot of SQL programming
  - Linux : familiar with the systems.




- Project 0: connect to the webserver follow day 0 slides





# Course Objective

- Evaluate the role of databases in computing systems
- Design and build applications using database management systems
- Compare and evaluate diverse database models
- Implement a consistent database
- Understand transaction processing concepts
- Understand relational algebra as a basis for database query languages

# About this class

- **NOT focus on**
  - A fancy web graphic user interface, but you need to design an online database application.
  - Data entry GUI, but you need to validate the input.
- **Do focus on**
  - Basic SQL and some advanced SQL
  - Database schema design
  - Data integrity
  - PL/SQL (Stored procedures)
  - Web Database application development
- Many discussions and coding
  - SQL
  - PHP MySQL in 3 tier architecture

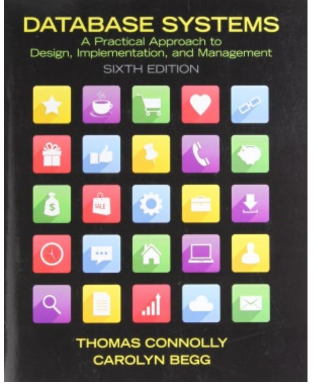
# About This Class – Cont'd

- Database is about coding.
- Best practices for coding
  - Style: Indents, comments, meaningful names, readable
  - Structure: reusable codes, module,
  - Organization: file/folders, separate code and data
  - Debugging: variables, break-point, maintenance
  - https://code.tutsplus.com/tutorials/top-15-best-practices-for-writing-super-readable-code--net-8118
- Not every slide will be covered -> self-study.
- Slides, assignments and materials will be shared
  - through OneDrive

# Course Information

- Pre-requisite:
  - CPS 2232 and Math 2110
  - Students without prerequisite(s) must withdraw from the class.
  - **Good programming skills**
- This is a hands-on course
  - Examples, project, homework, lab exercises
- Communicate through Kean email
- Requirements
  - Have Kean email account
  - Work on Linux servers
  - Use MySQL, PHP

# About This Course




- Required textbook: (Must have!)
  - Database Systems: A Practical Approach to Design, Implementation, and Management.
  - Author: Thomas Connelly and Carolyn Begg, 6th edition.
  - Addison-Wesley, 2014.
  - ISBN-13: 978-0132943260
- Need the textbook for exercises and lectures
- Bring your laptop every class for **exercise**.
- 30% of slides will NOT be covered -> **self-study**

# Important Policy

- All homework assignments, projects, and exams are expected to be completed by the students themselves.
- **No credit will be given for an assignment or homework that is copied – in part or in total – from another person**.
- **ChatGPT Usage:** You may use ChatGPT to assist with labs, projects, and homework, but **do not copy its answers.** Solve problems and write the code yourself to build problem-solving skills.
- **ChatGPT is strictly prohibited for quizzes, exams, and job interviews, as these assess your knowledge and skills, not those of ChatGPT.**

# Plagiarism Policy

- **Do NOT copy or modify other people's homework and lab assignments.**
  - Why pay tuition to copy other people's work!
  - Important to prepare quizzes, midterm and final.
  - Prepare for job interviews - internship and full-time job.
- **Any instance of plagiarism will result in a zero grade for the assignment!!!!**
  - **Samurai and friends are NOT excuses!**
  - **Discuss, but design/code by yourself!**
- **It is illegal to share your source codes**
  - **At school and at work**
  - **Same penalty!**

# Students are expected to …

- **Arrive class 5 minutes before class starts!!!**
  - **Self-study to catch up if missing a lecture or arriving late**
- **Study the slides before the class**
  - **Exercises** and **brainstorming** discussions in the class.
- **Spend more than 10 hours/week for average students**
  - **2.5 hours** /week in class
  - Outside of class: **7.5+ hours** /week on coding (& studying)
  - Not take more than 5 courses for full-time students
  - Not take more than 2 courses if working full-time
  - **60** available hours/week – 40 hours (work) = 20 hours left (study)
- **If you cannot commit 10 hours/week on this class, please take it later when you can.**
- **Bring a laptop every class**
  - A lot of exercises
- **NOT to post assignments/exams on the Internet**
- **Self-study topics for teaching presentations**

# Review syllabus

- **Class website**: https://vader.kean.edu/students
  - Check your current grade, get the permissions to our database server, view our class demo project, upload your photo…, and much more.

- Class materials at class OneDrive – **check NOW!**
- **Not to curve (also TRUE for seniors and graduate students!)**
  - **Please study, do the exercises and assignments.**
  - **Otherwise, please withdraw from this class!**
    - **Discuss, but design/code by yourself!**
- Class schedule (dates for assignments, exams)
  - Withdraw dates
  - Last Day to withdraw from courses with "W" grade
- Grading Policy and Grades
- Exam Policies
- Class Policies
- University Policies and Information

# Important Class Policy

- **Regardless of the senior status or job offers**, these two factors are not included in the grading criteria, students must diligently study to achieve good exam scores and submit functional assignments by the deadline to pass the class. Simply attending class is insufficient for students to successfully pass.
- **The final grade will not be curved, and instructors cannot manipulate or falsify the points you earned. This is a CS major elective course. If you cannot accept this policy, you should withdraw from this class immediately and take another major elective.**

# How to Succeed

- **Concentrate in classes!**
- Do lab exercises and all the assignments!
- Do homework and assignments as soon as it's been given.
- Read textbook and slides (before and after each class)
- **Self-study is VERY important (slides & textbook)**
  - Not all slides will be covered
  - Technology is changing faster than ever
- Study the course notes (before & after courses)
- Team work (Study group, but not cheating together)
- **PARTICIPATE!**
- **Seek help early, if necessary!**
- **Do your best!**

# Data Protection & Privacy

- You may become a Tech Support, DBA, System Administrator and Programmer who has special privileges to access other people's information
  - Respect people: privacy and civil rights
  - Federal privacy regulations: HIPPA and others
  - Violation: fine & criminal charge
- Learn how to protect yourself: user and customer
  - Change credit card number periodically
  - DO not put personal information on mail, cloud, web, social media, etc.

# Survey & Test

- Anyone not received my email last week?
- Follow day 0 slides to login to your Linux account
- Linux commands
  - How do you work on Linux? Which editor?
  - Transfer files between Windows and Linux
  - Commands: mkdir, cd, more, ps, mv
  - File permission modes
- PHP & HTML
  - Do you have a web page on any Linux server?
  - Familiar with PHP ASAP

# In this class, you will

- Have:
  - Access Linux web server
  - Access mysql database under server imc.kean.edu
  - Read only on database: CPS3740, dreamhome
    - Tables for homework and project.
  - Get some privileges on DB through class website
- Learn hands-on skills:
  - Unix commands
  - Database web application development
  - basic and advanced SQL statements
  - Access database through command-line and GUI

# Need help? Code Samurai

**Code Samurai :** https://yoda.kean.edu/~samurai/

- Need help?

(1) Check Samurai's skills on the 2nd page of this website, find the Samurai you will ask for help.

(2) Go to the 1st page to check his/her schedule and go to his/her session for help.

(3) Prepare before you ask for help. Resource is limited.

# Code Samurai program

- You should study and do the assignments first.
- Samurai is for trouble shooting, NOT responsible for teaching.
  - **They should not give you the answers directly!**
- Go to see Samurai if you cannot figure out the problems after working on the problems for 2 hours.
- Join Samurai team if you get an A or A-
  - Learn how to trouble shooting is very important in CS
  - Good for your resume

# Exercise – create your DB account

- Get the permission on the class website
  http://vader.kean.edu/students/
  -> bookmark the class website. You will need it very often.
  - Select and execute options 1 -> **create your own DB account**
  - Then select and execute options 2, 3, 4 to get the permissions on the databases, one at a time
  - (optional) Select and execute options 5 -> reset your password -> Memorize your password
- We will work on option 6 later

# Exercise – test your DB account

- Connect and login to your account (Refer to Day0 slides)
  - Windows: PuTTY
  - Mac: Terminal
- Connect to database, xxxx is your email id

  $ mysql -u xxxx -p -h imc.kean.edu
  - Type your password (student id). You can verify it on the class website
- Select the database dreamhome (case sensitive)

  mysql> use dreamhome;
- See the table Staff (case sensitive)
  - Do the following query and verify the results (6 rows)

  mysql> select * from Staff;

# Home Exercise

## Create project website folders in obi2 server

- **Create project website folders in obi2 server for your Homework, Lab assignments, and project.--** You can refer to Day0 slide for a step-by-step example -- '**Creating project website folders on the server**.'

- Useful commands under Linux command mode $
  - ls, cd, man, mkdir, mv, more, rm

**Please do these hands-on exercises in the class & complete it at home!!!**

# Chapter 1

**Introduction to Databases**

# Chapter 1 - Objectives

- Some common uses of database systems.
- Characteristics of file-based systems.
- Problems with file-based approach.
- Meaning of the term database.
- Meaning of the term Database Management System (DBMS).

# Chapter 1 - Objectives

- Typical functions of a DBMS.
- Major components of the DBMS environment.
- Personnel involved in the DBMS environment.
- History of the development of DBMSs.
- Advantages and disadvantages of DBMSs.

# Examples of Database Applications

- What data should be kept in the following examples:
  - Purchases from the supermarket
  - Purchases using your credit card
  - Borrowing books from the local library
  - Studying at Kean university

# Data saved in File-Based Systems

- Collection of application programs that perform services for the end users (e.g. reports).
- Each program defines and manages its own data.
- For example: using MS-Excel, or other systems.

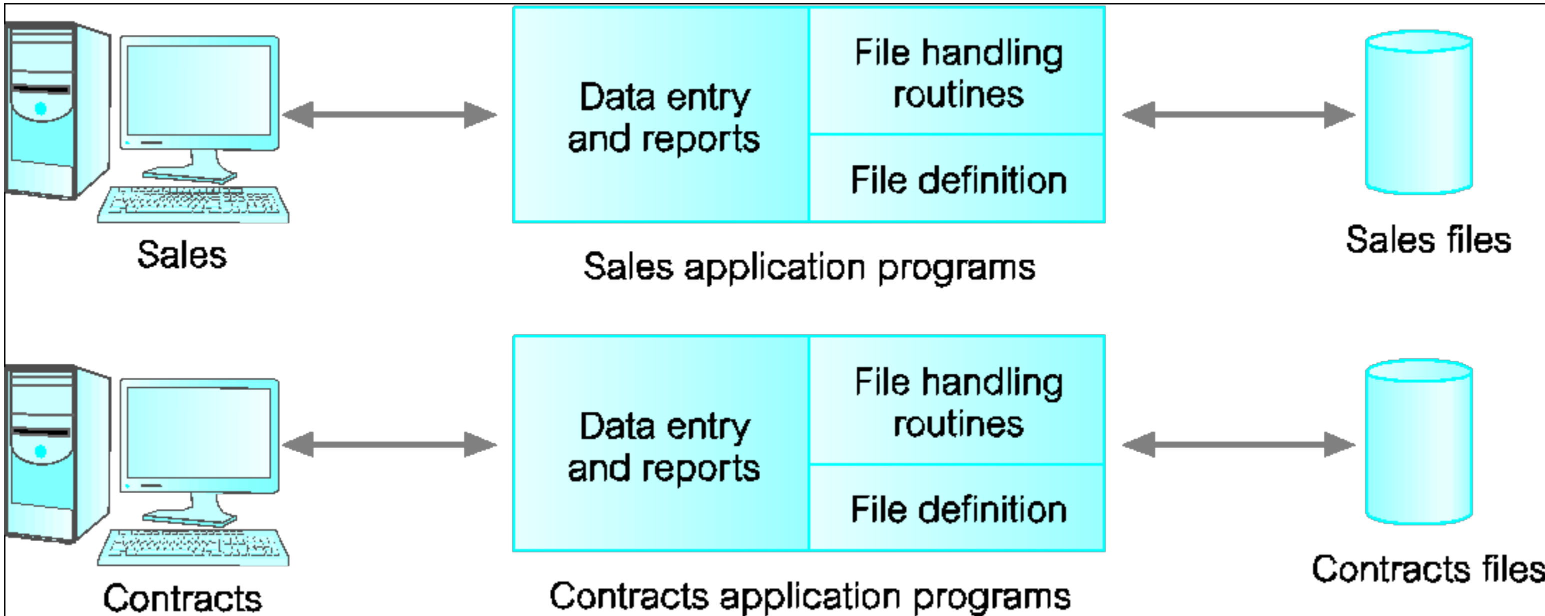
Sales Files

**PropertyForRent** (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
**PrivateOwner** (ownerNo, fName, lName, address, telNo)
**Client** (clientNo, fName, lName, address, telNo, prefType, maxRent)

Contracts Files

**Lease** (leaseNo, propertyNo, clientNo, rent, paymentMethod, deposit, paid, rentStart, rentFinish, duration)
**PropertyForRent** (propertyNo, street, city, postcode, rent)
**Client** (clientNo, fName, lName, address, telNo)

# File-Based Processing

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# File-Based example - Excel

- Excel – a file based data storage
  - Column, row
  - Data  type: integer, string, float
  - Cell variable: a1, b2, …
  - Cell functions: comparison, computation, …
  - Programming: macro using VB
  - Worksheet
  - Table functions: search, replace, AVG, SUM, …
  - Security: can lock data at file level
  - Compare two data cells in two different worksheet

# Limitations of File-Based Approach

What is the limitation for using Excel as a database?

**How to share?**

- **Separation and isolation of data**
  - Each program maintains its own set of data.
  - Users of one program may be unaware of potentially useful data held by other programs.

- **Duplication of data**
  - Same data is held by different programs.
  - Wasted space and potentially different values and/or different formats for the same item.

# Limitations of File-Based Approach

- **Data dependence**
  - File structure is defined in the program code.
- **Incompatible file formats**
  - Programs are written in different languages, and so cannot easily access each other's files.
- **Fixed Queries/Proliferation of application programs**
  - Programs are written to satisfy particular functions.
  - Any new requirement needs a new program.

# The Problems of using Excel

- Many different user/passwords on a file? -> NO
- More than one person write on the same file? ->NO
- For same user, only Read permission on worksheet A, but Read/Write permission on worksheet B? -> NO
- Either lock on entire file, or not lock at all.
- Cannot set constraint relationship between two or more worksheets
- Cannot force field(s) to be unique, primary key
- Cannot run as a server to serve other programs on different computers

# 1.3 Database Approach

- **Arose because:**
  - Definition of data was embedded in application programs, rather than being stored separately and independently.
  - No control over access and manipulation of data beyond that imposed by application programs.

- **Result:**
  - The database server.
  - Database Management System (DBMS).

# 1.3 Database

- Shared collection of logically related data (and a description of this data), designed to meet the information needs of an organization.

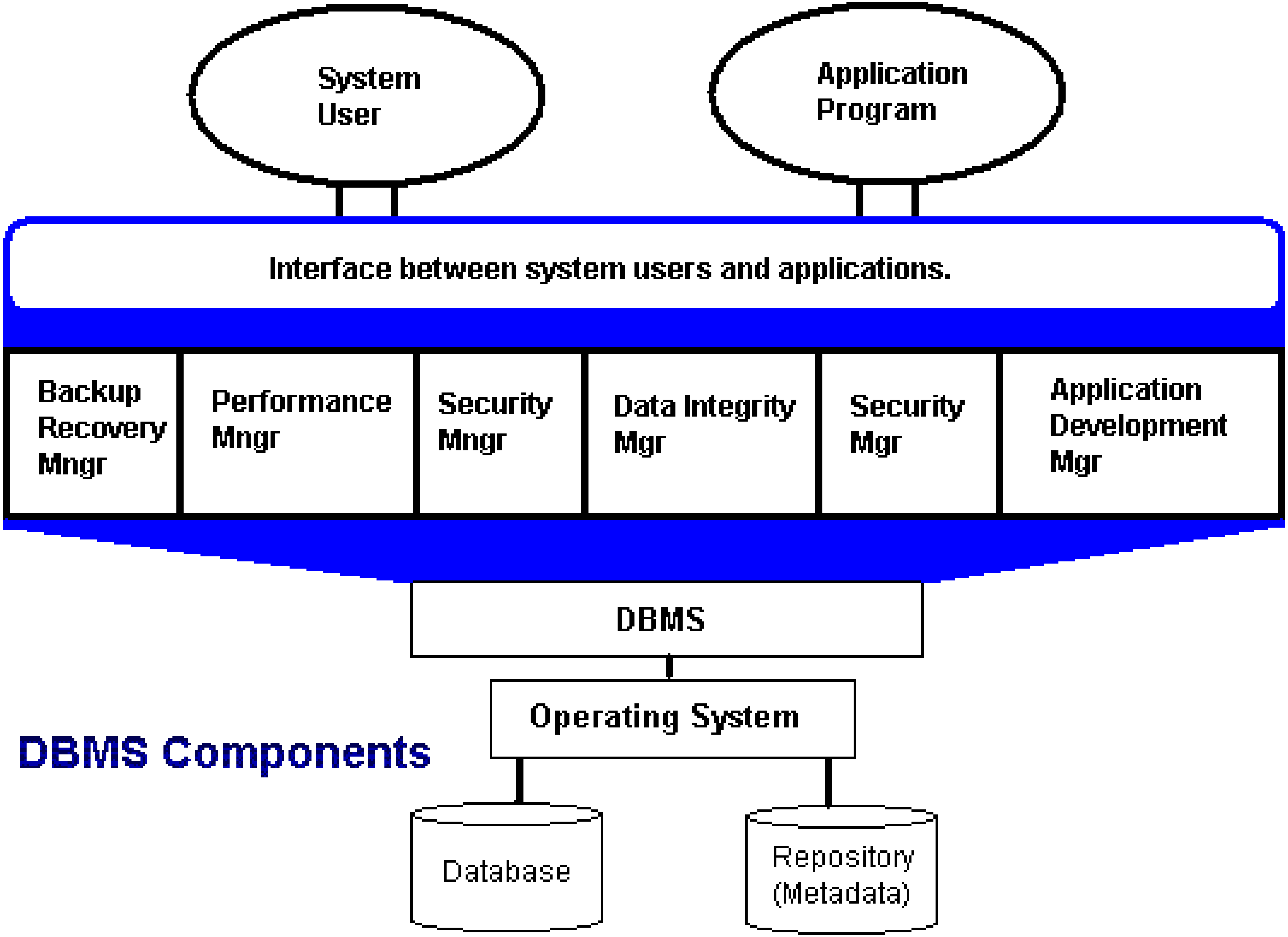
- System catalog (metadata) provides description of data to enable program–data independence.

- Logically related data comprises entities, attributes, and relationships of an organization’s information.

# 1.3 Database Management System (DBMS)

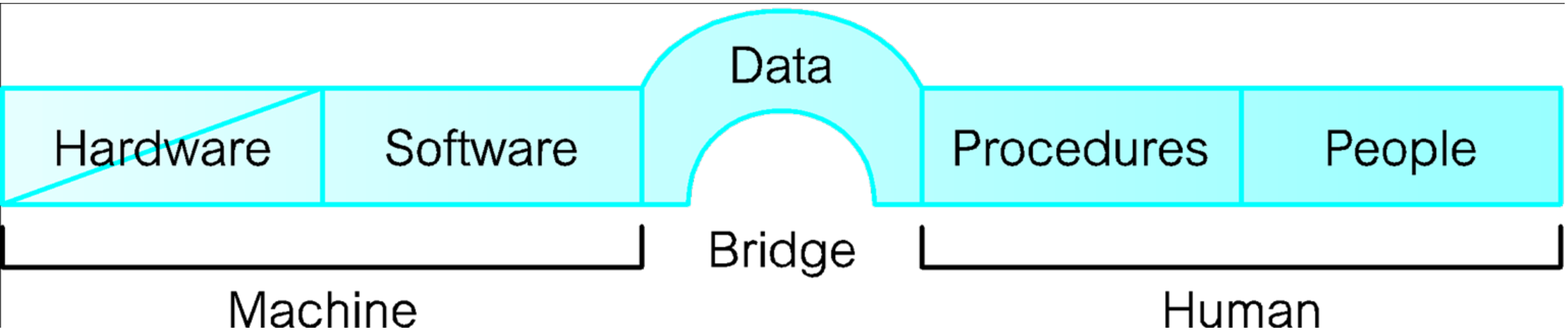
- A software system that enables users to define, create, maintain, and control access to the database.

- (Database) application program: a computer program that interacts with database by issuing an appropriate request (SQL statement) to the DBMS.




**DBMS Components**

# Components of DBMS Environment

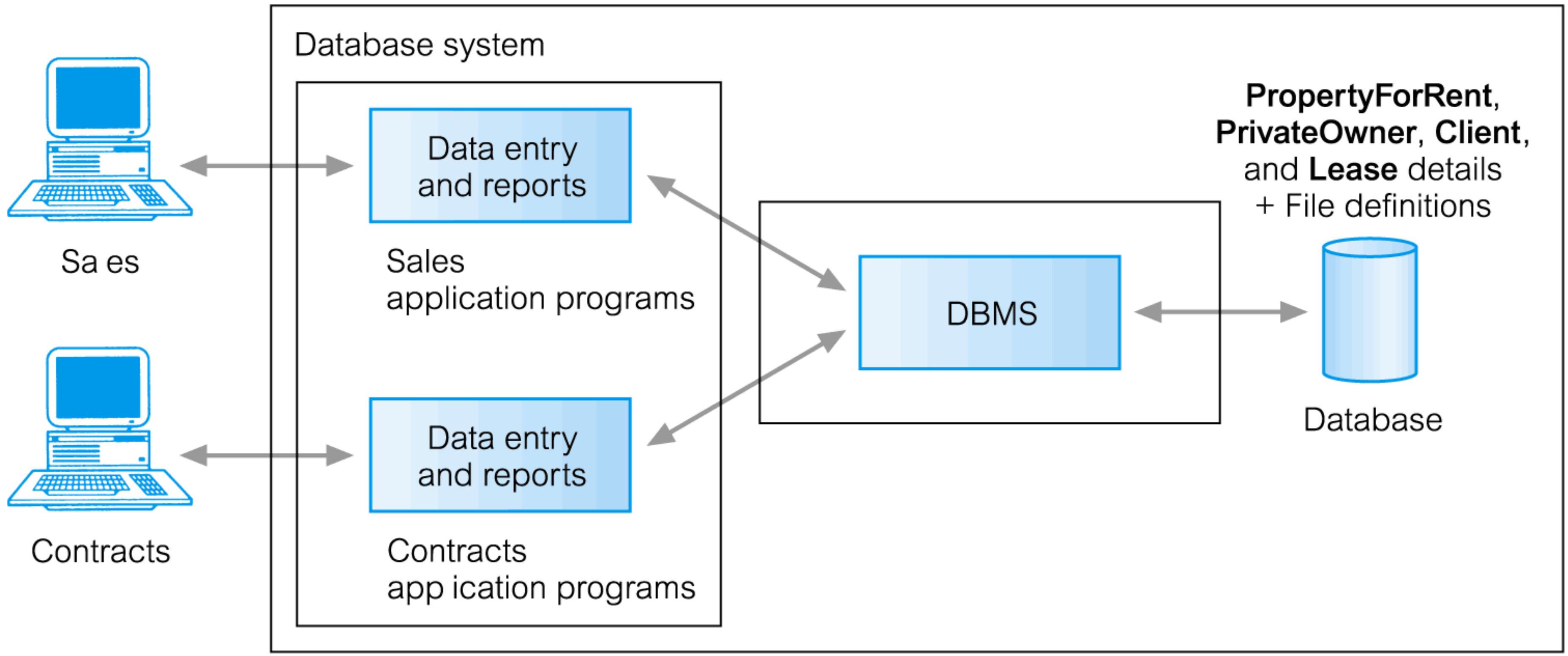



Why do you use your laptop or cellular phone?

Browse internet, check email, play games, watch videos, …

->These are all about data in different format.

# 1.3 Database Management

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**Lease** (leaseNo, propertyNo, clientNo, paymentMethod, deposit, paid, rentStart, rentF nish)

# 1.3 Components of DBMS Environment

- **Hardware**
  - Can range from a PC to a network of computers.
- **Software**
  - DBMS, operating system, network software (if necessary) and also the application programs.
- **Data**
  - Used by the organization and a description of this data called the schema.
- **Procedures**
  - Instructions and rules that should be applied to the design and use of the database and DBMS.
- **People**

# 1.4 Roles in the Database Environment

- **Data Administrator (DA)**
- **Database Administrator (DBA)**
- **Database Designers (Logical and Physical)**
- **Application Programmers**
- **End Users (naive and sophisticated)**

**Possible jobs/career**

# Functions of a DBMS

- A User-Accessible Catalog.
- Transaction Support.
- Concurrency Control Services.
- Data Storage, Retrieval, and Update.
- Recovery Services.
- Authorization Services.
- Support for Data Communication.
- Integrity Services.
- Services to Promote Data Independence.
- Utility Services.

Self study

# From Excel to Database - 1

- A file -> a database
- A worksheet -> A table
- Column -> attributes (name, age, grade, …)
- Row -> records
- Data type: integer, varchar, char, float, date, blob, …
- Working on multiple tables and database at same time.
  - Working -> join

Self study

# From Excel to Database

- Programming: Structure Query Language (SQL)
  - Functions: comparison, computation, …
  - User and data access control
  - Data level (Data Manipulation Language, DML): insert, delete, update
  - Table/structure level (Data Definition Language, DDL): create, alter, drop,
  - Privilege level (Data Control Language, DCL): grant, revoke
- Macro -> stored procedure
- All these above functions need a powerful system

Self study

# 1.6 Advantages of DBMSs - 1

- Control of data redundancy
- Data consistency
- More information from the same amount of data
- Sharing of data
- Improved data integrity
- Improved security
- Enforcement of standards
- Economy of scale

Self study

# 1.6 Advantages of DBMSs - 2

- Balance conflicting requirements
- Improved data accessibility and responsiveness
- Increased productivity
- Improved maintenance through data independence
- Increased concurrency
- Improved backup and recovery services

Self study

# 1.6 Disadvantages of DBMSs

- Why below are disadvantages?
  - Complexity
  - Size
  - Cost of DBMS
  - Additional hardware costs
  - Cost of conversion
  - Performance
  - Higher impact of a failure

Self study

# Home exercises

- Read and follow day0 slides
- Setup and create your project folder on the class webserver.
- Make your project URL working in a week.

Self study

# Reading Assignment

- 1.3 Database Approach
- 1.5 Roles in the Database Environment
- 1.6 Advantages and disadvantages of DBMS
- Google and study “client-server”

Self study

# Useful Links

- MySQL
  - http://dev.mysql.com/doc/refman/5.0/en/index.html
  - http://www.w3schools.com/sql/
  - http://www.mysqltutorial.org/
- PHP:
  - Tutorial http://www.php.net/manual/en/tutorial.php
  - PHP 5 http://www.w3schools.com/PHP/
- PHP MySQL
  - http://www.mysqltutorial.org/php-mysql/
  - http://www.w3schools.com/php/php_mysql_intro.asp
  - http://www.siteground.com/tutorials/php-mysql/

Self study

# Conclusion

- Read day0 slides
- Review chapter 1
- Create and test your database account
- Familiar with tools
  - Putty (Windows), terminal (Mac), FileZilla (Windows & Mac)

Self study